HODGE-DISPROOF: Dependent Type-Theoretic Schema and 31-Module Lean 4 Audit Framework for Rational Hodge Counterexample Candidates

Benjamin Stanley Frohman
ORCID: 0009-0006-7068-3718 | GitHub: BenFrohman/HODGE-DISPROOF
Dedicated 1-to-1 Zenodo DOI: 10.5281/zenodo.23653001

Earliest Commit Provenance: 2026-09-23 02:28:40 PM CDT (UTC-5)
Siloed Archival Edition: October 25, 2026 (America/Chicago CDT)

Abstract

Standalone 31-module Lean 4 formalization (‘Hodge/*.lean‘, ‘Main.lean‘, ‘EVIDENCE_LEDGER.md‘) specifying the exact dependent type-theoretic signature (D_bad, gamma_bad, forall z, cl(z) = gamma_bad -> False) required for any putative counterexample to the rational Hodge Conjecture, together with automated audit checks tracking which candidate geometric inputs are inhabited vs. open.

1 Scope, Scientific Honesty, and Non-Claim Boundary

Formalizes the type-theoretic verification schema and audit ledger for testing candidate counterexamples in Lean 4; explicitly records Clay status as OPEN and does not claim an inhabited unconditional disproof.

- **Repository:** `https://github.com/BenFrohman/HODGE-DISPROOF`
- **Dedicated 1-to-1 Zenodo DOI:** `10.5281/zenodo.23653001`
- **Earliest Git Commit SHA:** `974ac90ad5b6e7b134a51a37ea1b9131f8cc77c3` (2026-09-23 02:28:40 PM CDT (UTC-5))
- **License & Copyright:** Copyright © 2026 Benjamin Stanley Frohman. Licensed under the Apache License, Version 2.0.

2 Technical Overview & Repository Specification

Author: Benjamin Stanley Frohman (@BenFrohman)
Copyright: (c) 2026 Benjamin Stanley Frohman. Apache License 2.0.
License: Apache-2.0. See ‘LICENSE‘, ‘NOTICE.md‘, ‘AUTHORS.md‘, ‘COPYRIGHT‘.
Clay tag: ‘clay-statement-open‘. See ‘docs/CLAY_TAG.md‘.
Visibility: public. Forking allowed. Issues enabled. No repository rulesets.
https://github.com/BenFrohman/HODGE-DISPROOF
This repository records the Lean 4 **type** of a Hodge counterexample term.
It is **not** a Clay close. The rational Hodge conjecture remains **open**.
| Field | Value |
|—|—|

Visibility	public
Default branch	‘main‘
Forking	allowed
Issues	enabled
Rulesets	none
License	Apache-2.0
Clone	‘git clone https://github.com/BenFrohman/HODGE-DISPROOF.git‘
File	Role

3 Cryptographic Provenance & File Inventory

Every source artifact in `BenFrohman/HODGE-DISPROOF` is audited via `titan-audit` (`audit/TITAN_AUDIT_CERTI` and sealed under a deterministic SHA-256 Merkle tree via `sealguard` (`PROVENANCE_MERKLE_ROOT.json`) signed with Ed25519 key fingerprint `b428fb72676f1c52`.

Repository Path	Size	SHA-256 Prefix
AUTHORS.md	1141 B	5272a819c57d7ae2...
CHANGELOG.md	773 B	8c6ff5fa37a31dd9...
CITATION.cff	720 B	597ee27ad5255adb...
COPYRIGHT	407 B	f8414c79734515cb...
Hodge/AnalyticInterface.lean	2859 B	73f02439d9f8169f...
Hodge/Basic.lean	2234 B	3ebaefbd2ebf8fb9...
Hodge/ClayBlueprint.lean	2295 B	f1ec1b5d2a4d76bd...
Hodge/ClaySpec.lean	1162 B	6998297c71f31371...
Hodge/ClayStructure.lean	1955 B	57a68ea0b7b53dc4...
Hodge/CoeffRing.lean	1568 B	8347242ef3647111...
Hodge/ConjugateFiltration.lean	2090 B	7a947e863b0d29d6...
Hodge/Disproof.lean	2330 B	cf09c00cdbe9c9ff...
Hodge/EvenOddGadget.lean	2235 B	88280090520e08d7...
Hodge/FalseOfBadCl.lean	2266 B	693bb8734c14db1a...
Hodge/Fields123.lean	1409 B	68cfbd867eae2f09...
Hodge/FourfoldCase.lean	1490 B	260c79b96ed4c259...
Hodge/FourfoldMissType.lean	1163 B	93bca6ff5c30d831...
Hodge/GeometricDictionary.lean	1838 B	69bc48acdccb1785...
Hodge/GeometricHost.lean	3157 B	a9829d2fbdd679ea...
Hodge/HardLefschetz.lean	1821 B	7dfdb7a902149247...
Hodge/IntegratedGeometry.lean	1590 B	96adc5992e8501af...
Hodge/IntegratedSpec.lean	1936 B	e6ef5b7208065f97...
Hodge/LefschetzIteration.lean	1448 B	65c33c67280cb1b4...
Hodge/Lexicon.lean	1063 B	89569e16fbb4a32b...
Hodge/LinZhang.lean	975 B	1f09d6e1b93dfde3...
Hodge/LinZhangPolynomial.lean	791 B	e446a92e50a8e0bc...
Hodge/PeriodInterface.lean	1189 B	33c02d6a7a56856c...
Hodge/PrimitiveDecomposition.lean	1321 B	163393e411918b18...
Hodge/ProjectiveFourfoldSpace.lean	1163 B	712d77d9ac584a9f...
Hodge/RealProjectiveFourfold.lean	1831 B	d81220e40ca1938b...
Hodge/SchemeSpec.lean	1641 B	7fa822cae65e720d...
Hodge/ThomSebastiani.lean	1024 B	36a3f0ec6ab0079c...
Hodge/Typecheck.lean	1500 B	079be03225284a28...
Hodge/VF.lean	812 B	190f8621e6ef5475...
Hodge/X5.lean	1411 B	2c0f85fb8bc80ffe...

Attribution & Licensing Notice: Copyright © 2026 Benjamin Stanley Frohman (ORCID: 0009-0006-7068-3718). Licensed under the Apache License, Version 2.0.